Unit - 5
Alkali and Alkaline Earth Metals

EVALUATION

I Choose the best answer.

1. (c) Density: Li < K < Na < Rb < C
2. (a) Li^+ has minimum degree of hydration among alkali metal cations
3. (d) none of these
4. (b) Li
5. (c) kerosene
6. (a) superoxide and paramagnetic
7. (c) Potassium carbonate can be prepared by solvay process
8. (b) Magnesium
9. (b) MI < MBr < MCl < MF
10. (a) Castners process
11. (c) $Ca(CN)_2$
12. (a) $MgCl_2$
13. (a) p-2, q-1, r-4, s-5, t-6, u-3
14. (d) both assertion and reason are false
15. (a)
16. (b) $MgCO_3$ > $CaCO_3$ > $SrCO_3$ > $BaCO_3$
17. (c) Its salts are rarely hydrolysed
18. (c) milk of lime
19. (b) $NaHCO_3$
20. (b) $Ca(OH)_2$
21. (a) Ca^{2+} ions are not important in maintaining the regular beating of the heart.
22. (b) CaF_2
23. (a) $CaSO_4.2H_2O$
24. (b) CaNCN
25. (d) Li_2CO_3

Keys to multiple choice questions:

1) Option (c)

Potassium is lighter than sodium

(Refer table 5.3)

The correct order of density is

Li < K < Na < Rb < Cs

0.54 < 0.86 < 0.97< 1.53< 1.90(in g cm^{-3})

2) Option (a)

Li^+ has maximum degree of hydration among alkali metal cations.

$Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$

3) All these compounds reacts with alkali metals to evolve hydrogen gas.

(d) none of these.

4) hydration energy of Li^+ is more and hence Li^+ is stabilized in aqueous medium.

(b) Li

5) (c) Kerosene

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